

Neural Shield AI

Real-time scam and fraud detection —
across text, images, URLs, QR codes, and voice.

Startup Overview · 2025

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Executive Summary

Neural Shield AI is a multi-modal fraud detection platform that analyzes digital content — text messages, images, URLs, QR codes — for scam signals in real time. It is built for a world where scammers have industrialized their operations and individuals, banks, and platforms lack the tooling to keep up.

The platform combines a deterministic rules engine with large language model reasoning (Claude by Anthropic) to deliver verdicts at low cost and high speed. Neural Shield AI is production-ready for text, URL, UPI, phone, screenshot, and QR analysis today, with voice pipelines in active development.

The target customer is anyone who handles user-generated content at scale: fintechs, telecom providers, e-commerce platforms, and government bodies overseeing consumer protection. The platform is live as a web app, a Chrome browser extension, and a REST API.

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Problem Statement

The scale of online fraud has crossed a threshold where manual awareness campaigns and reactive blacklists cannot keep up. India alone records over 7,000 cyber fraud cases every day. Globally, scam-related losses exceeded \$1 trillion in 2023. Attack vectors have expanded well beyond email phishing: scammers now operate across WhatsApp messages, fake payment screenshots, QR codes embedded in physical flyers, and AI-generated voice calls impersonating bank officials and government agencies.

Existing tools are fragmented and modality-specific. Spam filters catch known keywords but miss context. Reverse image search requires deliberate user effort and fails on screenshots. URL scanners miss domains registered hours before a campaign launches. No single product handles all modalities, which means scammers simply shift to the channel with the least friction.

The people who suffer most are not always the least careful — they are the least equipped. Elderly users are disproportionately targeted by authority impersonation. Rural populations encounter QR-code payment scams at local markets. SMEs are defrauded through fake invoice URLs and UPI collect requests. Financial platforms absorb fraud losses when their customers are deceived before a transaction reaches the bank.

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Our Solution

Neural Shield AI is a unified detection platform that routes any piece of digital content — a WhatsApp message, a QR code from a flyer, a screenshot of a payment screen, a suspicious link — through a multi-layered analysis pipeline and returns a structured verdict within seconds.

The core insight is that scams leave signals across multiple modalities simultaneously. A scam SMS contains urgency language, links to a newly registered domain, and a QR code that encodes a collect-request UPI ID. By analyzing all of these together, Neural Shield AI catches what single-modality tools miss. The platform returns a trust score, a risk level, a confidence score, and a plain-language

explanation — not just a binary flag.

The platform is exposed in three ways: a web app for end users, a Chrome browser extension for real-time link and page checking, and a REST API for enterprise integration. All three channels share the same scoring engine, ensuring consistent verdicts across every touchpoint.

What It Can Do

Production — Live Today

Text / SMS / WhatsApp	Detects credential requests (OTP, PIN, CVV), urgency threats, lottery and prize scams, loan fraud, job scams, brand impersonation, and UPI collect-request patterns.
Email Analysis	Checks sender domain against the claimed brand, validates SPF and DMARC DNS records, flags free-email addresses impersonating companies (e.g. hdfc.support@gmail.com).
URL Scanning	RDAP domain age, TLS certificate checks, redirect chain expansion, Google Safe Browsing, URLHaus, PhishTank, VirusTotal v3, Spamhaus DBL/ZEN, AbuseIPDB, DNS TTL anomaly and sinkhole IP detection — 11 collectors in parallel.
UPI / Payment ID	Unknown PSP detection, brand-PSP mismatch on intent URIs, suspicious handle keywords (prize, kyc, refund, verify), collect-request language detection.
Phone Number Analysis	Premium-rate prefix detection (900x, 140x), suspicious digit sequences, non-Indian numbers paired with claims of Indian bank or government authority.
Screenshot OCR	Extracts text from images via Tesseract OCR and runs it through the full text pipeline — catches fake payment screenshots and fraudulent app UIs sent as 'proof.'
QR Code Decoding	Decodes QR payloads and routes them to the URL or UPI scanner — covers codes embedded in images, flyers, and forwarded screenshots.
Chrome Extension	Real-time analysis of the active tab's URL and embedded links, popup verdict and signals panel, Supabase auth, offline rule-based checks for zero-latency feedback.

In Development

■ Voice / Call Scam Detection	Real-time transcription of phone calls + LLM analysis for authority impersonation and social engineering patterns.
■ Vernacular Language Support	Detection in Hindi, Tamil, Bengali, and other Indian regional languages — keyword sets are additive to the rules engine, not a model retraining.
■ Enterprise Batch API	High-volume scanning endpoint with SLA guarantees, dedicated rate limits, and per-organisation reporting dashboards.

How We Built It

Layer 1 — Deterministic Rules Engine

Fast, zero-cost structural checks run first. Pattern matching for scam keywords, brand impersonation regex, UPI PSP validation against a curated allowlist, phone prefix tables mapped to TRAI data, and domain age thresholds. These checks run in under 1 millisecond and handle the majority of clear-cut cases without any external API calls.

Layer 2 — Threat Intelligence Collectors

11 parallel network collectors run under a 4-second global budget: RDAP, TLS, Google Safe Browsing, URLHaus, PhishTank, OpenPhish, VirusTotal v3, Spamhaus DBL and ZEN, AbuseIPDB, DNS TTL/sinkhole detection, and SPF/DMARC. All collectors are fail-open — a timeout lowers the confidence score but never crashes the scan. Results are cached (1–4 hours per source) to avoid redundant API calls on repeated targets.

Layer 3 — Reputation Engine

Community reports aggregated in Supabase. Entities reported by multiple users accumulate a weighted score. A score above threshold triggers a hard override to maximum risk. Clean entities with verified intel receive a trust-boosting negative signal.

Layer 4 — LLM Explanation (Claude by Anthropic)

Only after the deterministic pipeline has produced a final risk score does Claude enter — and only to write a plain-language explanation of that verdict. Claude never generates the score, the risk level, or the probability. This separation ensures the scoring is reproducible and auditable; the LLM's role is communication, not decision-making.

Routing logic: inputs are classified by type (URL, email, phone, UPI, text, image, QR) and dispatched to the relevant sub-pipeline. Images go through OCR first; QR codes are decoded first. Entity extraction from message bodies surfaces embedded URLs, UPI IDs, and phone numbers, each of which runs its own sub-pipeline. Compute is spent only where the input warrants it.

Tech Stack

Frontend	Next.js 16, Tailwind CSS v4, React Query, Recharts, TypeScript
Backend	Node.js, Express 4, TypeScript, Zod, Winston, Helmet, rate-limiting
AI / ML	Claude (Anthropic) via OpenRouter; Tesseract.js (OCR); jsqr (QR decoding)
Threat Intel	Google Safe Browsing, VirusTotal v3, Spamhaus DBL+ZEN, AbuseIPDB, URLHaus, PhishTank, OpenPhish, custom RDAP / TLS / DNS collectors
Database	Supabase PostgreSQL + Row Level Security, Supabase Auth, Supabase Edge Functions (Deno)
Infrastructure	Vercel (frontend + backend serverless), Supabase Cloud

Payments	Razorpay via Supabase Edge Function; server-side HMAC verification; backend never sees the Razorpay secret
Extension	Chrome Manifest V3, service worker, chrome.storage.local

Architecture Overview

Every scan request follows the same deterministic path. No branching logic produces inconsistent results across environments or deployments.

Input	Request arrives via web UI, Chrome extension, or REST API with a bearer token or API key.
Auth	JWT validated against Supabase Auth. API key hashed and matched against the api_keys table. Daily limits enforced via SECURITY DEFINER RPC — the backend holds no service-role key.
Normalize	Entity canonicalized: URLs stripped of tracking params and punycode-decoded, emails split into local/domain parts, phones normalized to E.164, UPI VPAs extracted from intent URIs.
Rules Engine	Deterministic checks in under 1ms: keyword regex, brand impersonation, UPI PSP lookup, phone prefix tables. Highest-reliability signals.
TI Collectors	Up to 11 collectors run concurrently under a 4-second budget. Each returns signals or an empty array — never an exception. Coverage and failure rates feed confidence.
Reputation	Community report aggregate queried. Result: hard override, abuse signal, prior verdict, trust boost, or nothing.
Risk Engine	Signals accumulated with per-category caps and source-tier multipliers. riskScore = sum(capped positives) + sum(negatives), clamped to 0–100.
AI Explanation	Claude receives entity type, risk level, confidence, and signal list. Returns summary, recommendation, and scam-type label. No numbers generated by AI.
Response	scamProbability, trustScore, riskLevel, confidence, signals[], flags[], explanation, engineVersion, processingTimeMs — scan persisted under user's RLS context.

Cost & Efficiency

The tiered pipeline was designed from day one with cost-per-scan as a first-class constraint. The deterministic rules engine handles the bulk of scans at effectively zero marginal cost — no API calls, no token spend.

- TI API results are cached 1–4 hours per source, eliminating redundant calls on repeated targets.
- Claude is called only for the explanation step — token usage is ~200–400 tokens per scan.
- Estimated cost at scale: under \$0.005 for a text or URL scan; under \$0.02 for image + OCR + TI.
- 80%+ of scans are resolved by the rules engine alone, with no external API calls.
- TI collectors are gated by entity type: VirusTotal and AbuseIPDB are never called for phone-only scans.

How We Scale

Stateless Backend	Express runs as Vercel serverless functions today; trivially portable to Cloud Run or ECS for higher throughput. The Dockerfile is already in the repo.
Database	Supabase PostgreSQL with RLS at the DB layer. Horizontal read replicas require no application code changes.
TI Cache	In-memory LRU cache with configurable TTL per source. Redis is a drop-in replacement when single-instance cache becomes a bottleneck.
New Modalities	The collector interface is plug-in: a new TI source is ~80 lines implementing <code>appliesTo()</code> , <code>isConfigured()</code> , and <code>collect()</code> . The orchestrator requires no changes.
Language Expansion	Rules engine uses regex patterns. Adding Hindi, Tamil, or Bengali keyword sets is additive — no architecture changes, no model retraining.
Enterprise	API key auth is already shipped. Business-tier customers receive higher daily limits, dedicated rate limits, and per-key usage reporting.
Partnerships	TRAI DIP, caller ID databases, CERT-In feeds, and bank fraud-desk APIs slot in as new TI collectors — the integration surface is well-defined.

Target Market

Individuals (B2C)	Urban and semi-urban smartphone users who receive scam messages daily. Entry via web app and Chrome extension.
Fintechs & Banks	Screen incoming payment requests, validate UPI IDs, flag suspicious transactions before processing. REST API.
Telecom Providers	Screen SMS content at the carrier level before delivery. High-volume batch API.
E-Commerce Platforms	Seller fraud, fake payment screenshots used to claim non-payment. Screenshot OCR and URL scanning.
Government & Regulatory	CERT-In, TRAI, consumer protection agencies — bulk threat intelligence and incident reporting dashboards.
Elderly & Vulnerable	White-label via NGOs, banks' senior citizen programs, or telecom CSR — simplified UI, vernacular language.

Market opportunity: India's cybersecurity market is projected at \$3.5B by 2027. The global fraud detection market reaches \$63B by 2029. Neural Shield AI targets the underserved real-time content screening layer — the gap between enterprise SIEM tools and individual awareness campaigns.

Roadmap

Near-Term (next 6 months)

- Complete voice/call scam detection: real-time transcription + LLM analysis for authority impersonation
- Launch vernacular language support: Hindi, Tamil, Bengali keyword sets
- Enterprise batch API GA with SLA guarantees and dedicated infrastructure
- TRAI DIP integration for caller ID enrichment
- Android SDK for in-app scam detection by third-party apps

Medium-Term (6–18 months)

- Deepfake image and video detection module
- Financial document fraud detection: fake bank statements and KYC documents
- Carrier-level SMS screening partnerships via telco API
- White-label product for banks' customer-facing mobile apps
- Regulatory reporting dashboard for CERT-In and RBI Ombudsman integration

Closing

Neural Shield AI is being built because scammers have industrialized — and the people protecting consumers have not kept up. The platform already runs a production trust engine with 11 threat intelligence collectors, community reputation signals, and a deterministic scoring pipeline that keeps AI in an explanation role rather than a black-box decision role. Every weight, every threshold, and every signal is versioned and auditable. The next phase is expanding the modalities, the languages, and the distribution channels. The goal is a world where anyone — regardless of digital literacy — can verify before they trust.